SRI MANAKULA VINAYAGAR
ENGINEERING COLLEGE
(AN AUTONOMOUS INSTITUTION)

DEPARTMENT OF COMPUTER AND COMMUNICATION ENGINEERING

EchoTrail — Personal Digital-Footprint Exposure Auditor

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INTRODUCTION

- People share large amounts of information through social media, professional platforms, public repositories, and online services.
- Individually harmless pieces of public information can become sensitive when they are combined across multiple platforms.
- Existing privacy and breach-checking tools generally examine only one source or one type of exposure at a time.
- Users therefore lack a consolidated understanding of **what a stranger or attacker could infer from their public digital footprint**.
- Publicly available information such as location tags, workplace details, usernames, repository activity, and exposed email addresses can contribute to social-engineering and doxxing risks.
- **Proposed Solution — EchoTrail:** A privacy self-assessment platform that aggregates consented public information and performs **cross-platform correlation-risk analysis** to generate an understandable exposure report and remediation checklist.

PROBLEM STATEMENT

- Public digital information is distributed across multiple platforms and is difficult for users to evaluate collectively.
- Existing privacy tools are largely **siloed**, providing platform-specific or breach-specific information.
- Users may not realize how apparently unrelated public information can be combined to reveal sensitive patterns.
- Location information, workplace details, public usernames, repository activity, and online routines may collectively increase exposure.
- There is no simple self-audit system that explains **what an unknown person could potentially infer** from a user's public footprint.
- Users lack a prioritized mechanism to identify high-risk exposures and determine which privacy changes should be made first.
- Manual self-auditing is time-consuming and requires users to understand OSINT and privacy-risk concepts.

LITERATURE SURVEY

TITLE	JOURNAL / YEAR / REF. NO.	METHODOLOGY	DRAWBACK
Have I Been Pwned	Existing Platform	Checks whether an email address appears in known data breaches.	Focuses mainly on breach exposure; does not provide cross-platform correlation analysis
Google Privacy Checkup	Existing Platform	Allows users to review selected privacy and account settings.	Limited to the user's Google ecosystem.
Facebook Privacy Checkup	Existing Platform	Reviews audience and privacy settings for Facebook features.	Does not analyze exposure across external platforms
Maltego	OSINT Platform	Performs relationship and entity analysis across data sources.	Primarily designed for investigations rather than personal privacy self-auditing
Proposed: EchoTrail	Proposed System, 2026	Combines consented public-profile information, breach checks, NER extraction and correlation-risk analysis	Requires careful privacy controls, false-positive handling and ethical data-use safeguards

EXISTING SYSTEM

- **Individual Privacy Checkups:** Platforms provide privacy settings and security recommendations within their own ecosystems.
- **Breach Checking Services:** Tools such as Have I Been Pwned allow users to determine whether their email addresses appear in known breaches.
- **Social Media Privacy Controls:** Users can manually review profile visibility, posts, location information, and account settings.
- **OSINT Investigation Tools:** Platforms such as Maltego can discover relationships between publicly available entities and information.
- **Manual Digital-Footprint Auditing:** Users can search their name, username, email address, and other identifiers across different websites.
- **Fragmented Exposure Assessment:** Information from different platforms generally remains disconnected rather than being analyzed as one combined digital footprint.

DISADVANTAGES OF EXISTING SYSTEM

- **Data Fragmentation:** Public information is distributed across multiple platforms and services.
- **No Cross-Platform Correlation:** Existing privacy tools generally do not connect information from different platforms to identify combined exposure.
- **Limited Risk Context:** Users may be told that information is public without understanding what could potentially be inferred from it.
- **Manual Analysis:** Users must independently search and compare their online information.
- **No Prioritized Remediation:** Existing tools may provide recommendations but do not necessarily prioritize actions according to combined exposure risk.
- **Investigator-Oriented Tools:** Many OSINT platforms are designed for professional investigations rather than personal privacy self-assessment.
- **No Exposure History:** Users generally lack a simple way to track whether their digital exposure improves or increases over time.

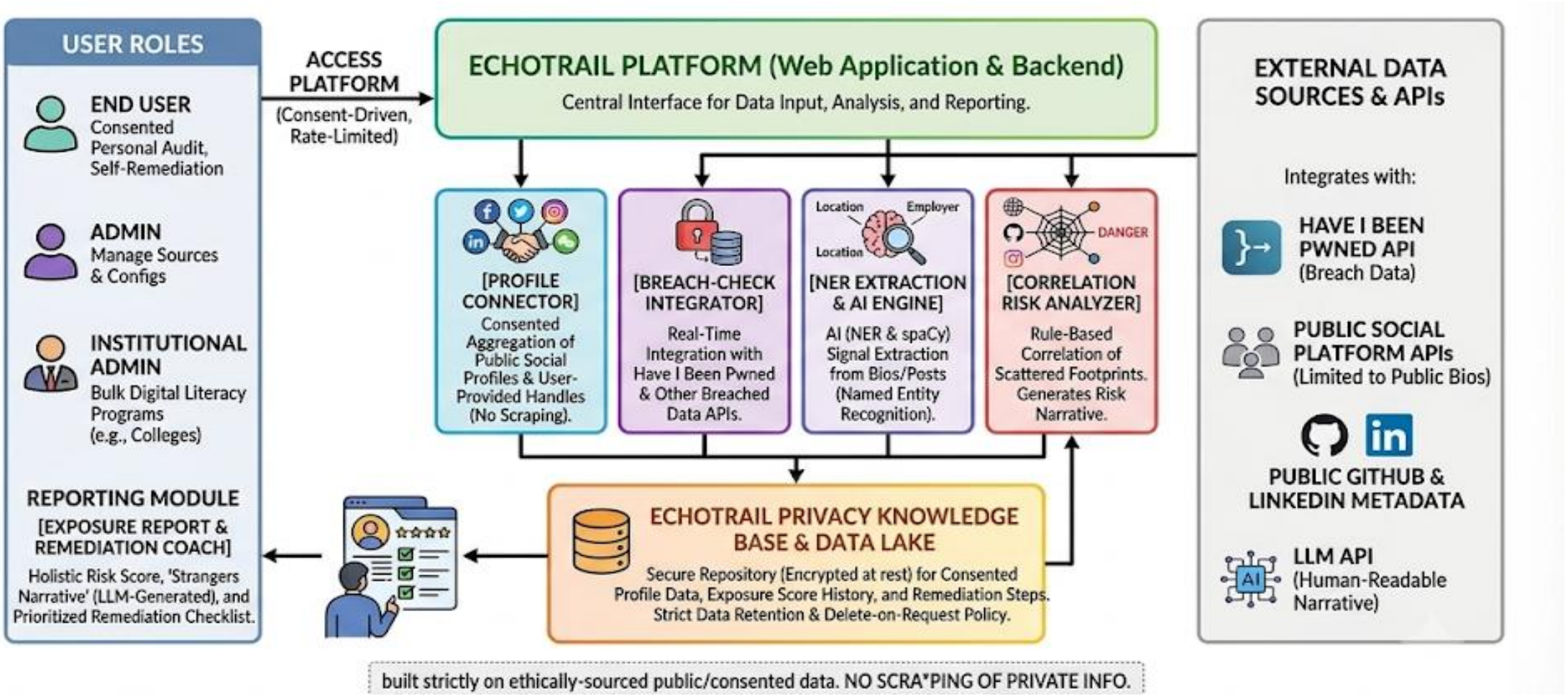
OBJECTIVES

- **Unified Exposure Assessment:** Aggregate consented public digital-footprint information into a single dashboard.
- **Breach Monitoring:** Check user-provided identifiers against supported breach-data sources.
- **NER-Based Extraction:** Identify entities such as locations, organizations, workplaces, and other exposure signals from public text.
- **Correlation-Risk Analysis:** Identify relationships between apparently separate pieces of public information.
- **Exposure Scoring:** Generate category-wise risk indicators for location, workplace, contact information, routine, and related exposure.
- **Remediation Guidance:** Provide a prioritized checklist explaining practical actions users can take to reduce exposure.
- **Progress Tracking:** Maintain exposure-score history to help users monitor improvements after privacy changes.

PROPOSED SYSTEM

- **Centralized Privacy Dashboard:** Provides a unified view of the user's digital-footprint exposure.
- **Profile Connector:** Accepts user-provided handles or consented public-profile information without accessing private accounts.
- **Breach-Check Integration:** Connects supported breach-check services to identify exposed user identifiers.
- **NER Extraction Engine:** Extracts relevant entities such as locations, organizations, employers, and routine-related signals from public information.
- **Correlation Risk Analyzer:** Connects information from different sources to identify potentially sensitive combinations.
- **Dynamic Exposure Scoring:** Calculates category-wise exposure levels based on identified signals and correlation rules.
- **Remediation Coach:** Generates prioritized recommendations for reducing unnecessary exposure.

SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE ECHOTRAIL

1. USER & ACCESS CONTROL

- **End User:** Personal privacy audit & self-remediation
- **System Admin:** API keys, breach feeds & system management
- **Institutional Admin:** College/organization-level digital-literacy programs
- **Consent & Anti-Abuse Gateway:** Authentication, consent verification & rate limiting

2. CORE PROCESSING PLATFORM

- **Profile Connector:** User-provided handles & public profile data
- **Breach-Check Integrator:** Breach verification through supported APIs
- **NER & AI Engine:** spaCy/HuggingFace for location, employer & routine extraction
- **Correlation Risk Analyzer:** Connects scattered footprints to identify potential threat vectors

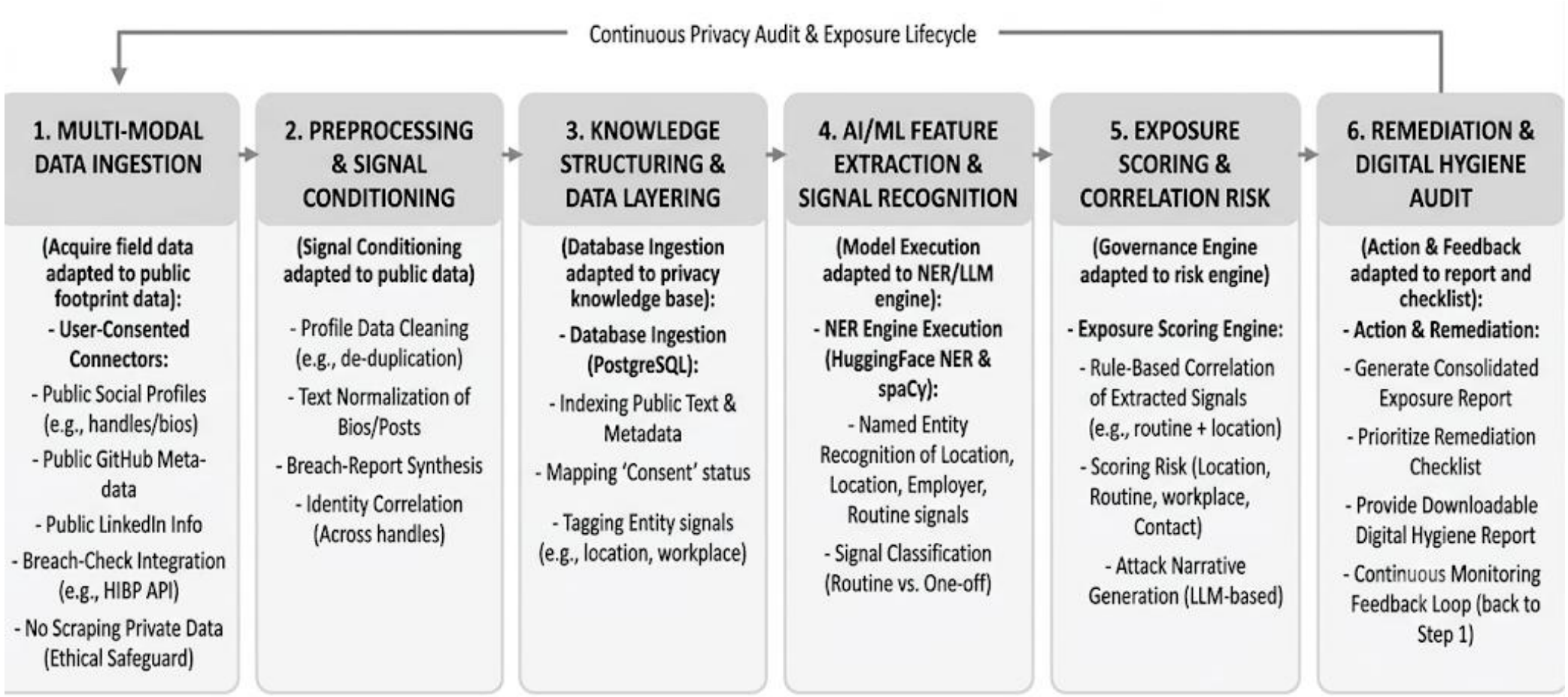
3. DATA LAKE & PRIVACY STORAGE

- **Encrypted PostgreSQL Repository:** Profile data, entity logs & exposure history
- **Privacy Controls:** Encryption at rest, short-term retention & **Delete All My Data**
- **GDPR-style data governance**

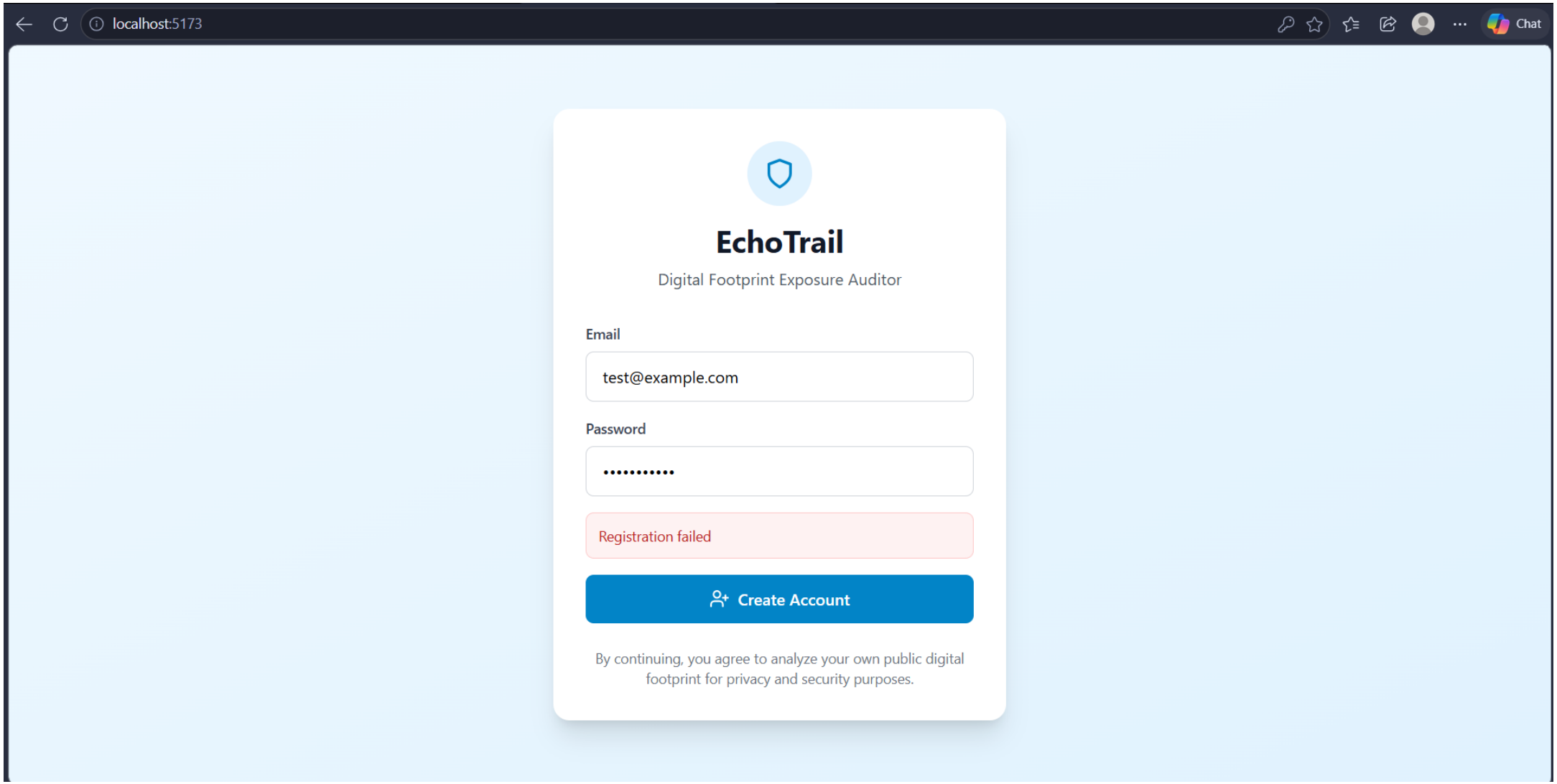
4. EXTERNAL SERVICES & REPORTING

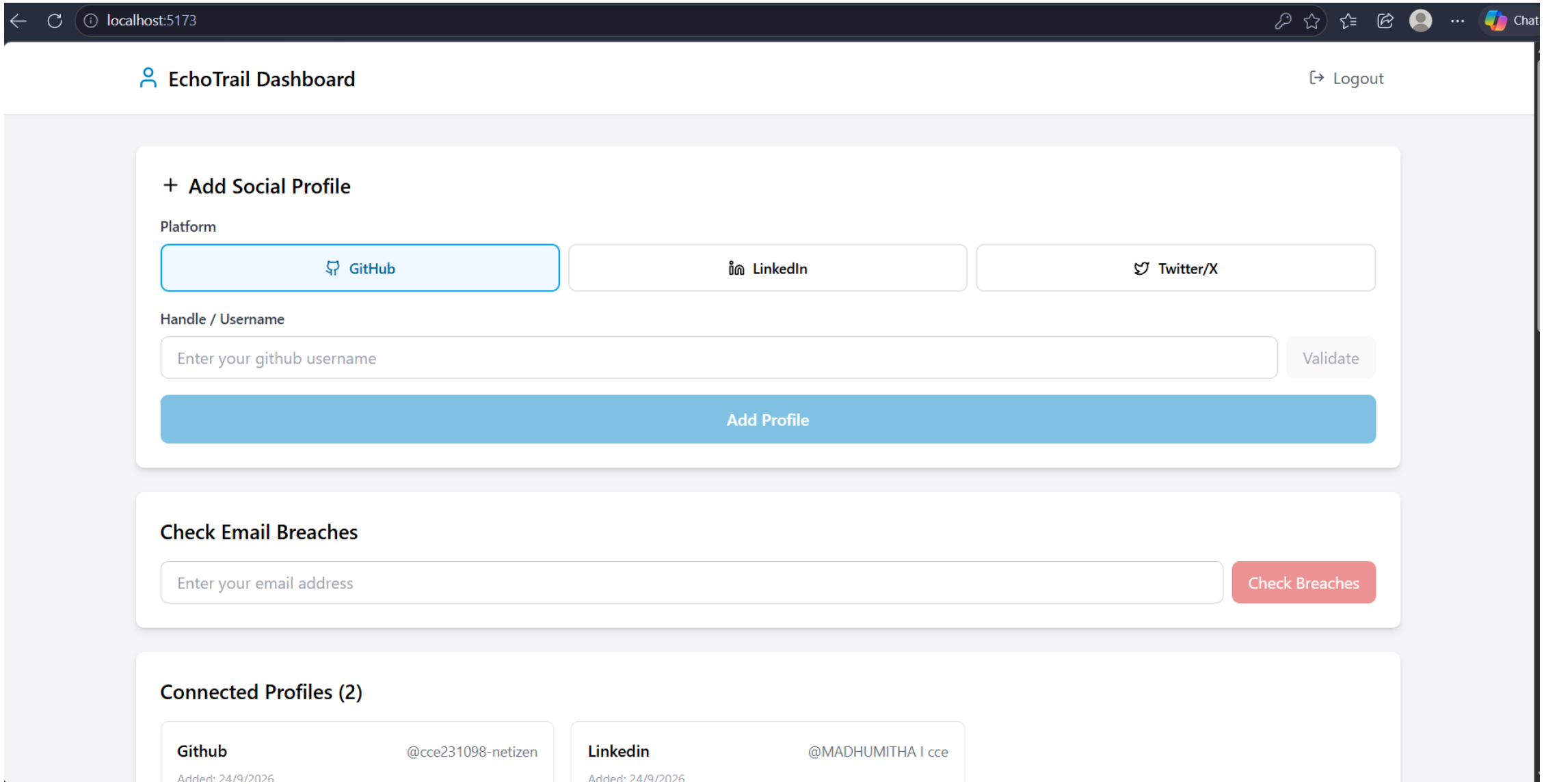
- **External APIs:** HIBP, GitHub public metadata & supported social APIs
- **LLM Reporting:** Generates **"What an Attacker Learns About You"** narrative
- **Remediation Coach:** Provides prioritized privacy-improvement checklist

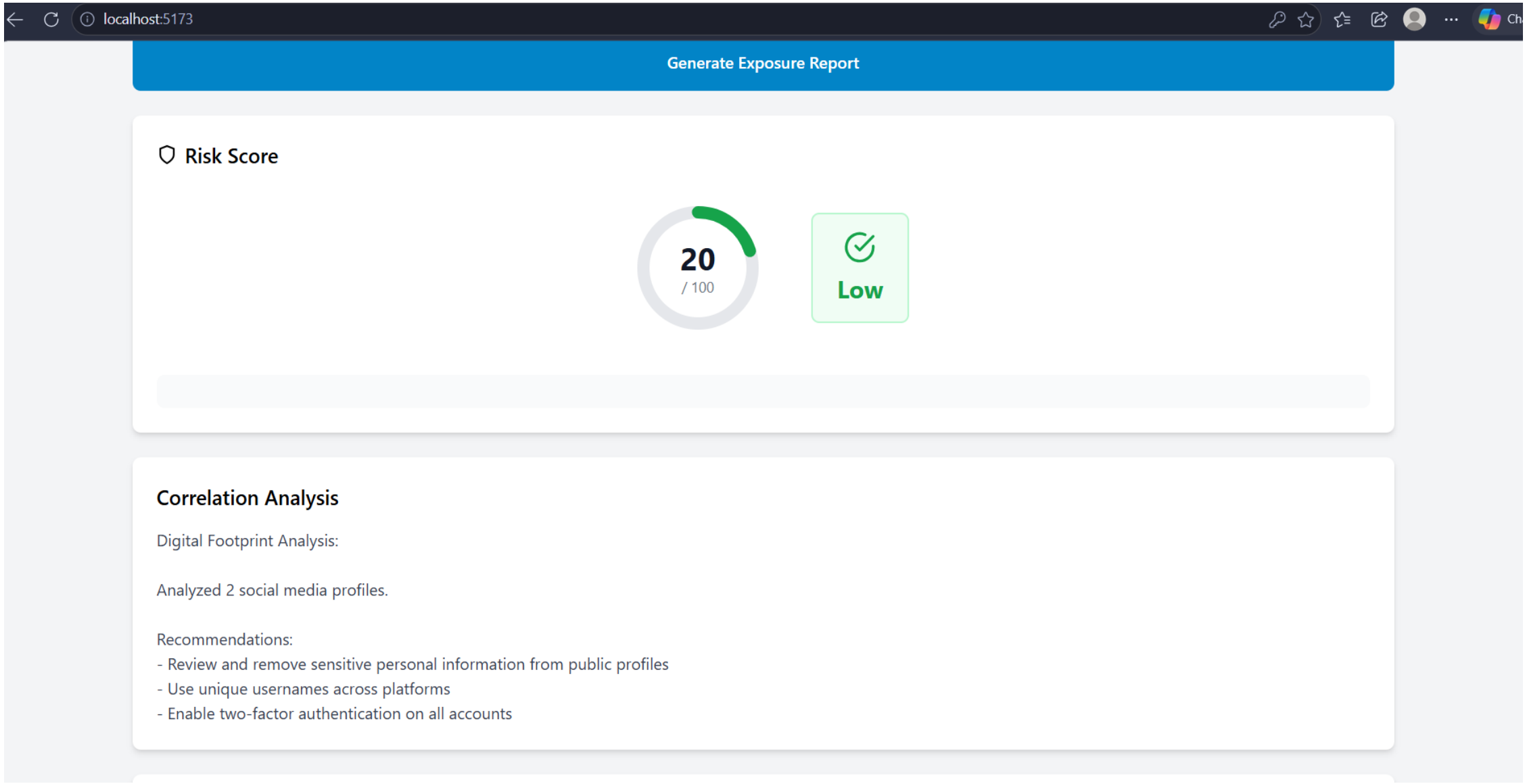
PROPOSED METHODOLOGY



SCREENSHOTS




localhost:5173
Chat
EchoTrail Dashboard
Logout
+ Add Social Profile
Platform
GitHub
LinkedIn
Twitter/X
Handle / Username
Enter your github username
Validate
Add Profile
Check Email Breaches
Enter your email address
Check Breaches
Connected Profiles (2)
Github
@cce231098-netizen
Added: 24/9/2026
Linkedin
@MADHUMITHA I cce
Added: 24/9/2026


localhost:5173
Generate Exposure Report
Risk Score
20
/ 100
Low
Correlation Analysis
Digital Footprint Analysis:
Analyzed 2 social media profiles.
Recommendations:
- Review and remove sensitive personal information from public profiles
- Use unique usernames across platforms
- Enable two-factor authentication on all accounts

Correlation Analysis

Digital Footprint Analysis:

Analyzed 2 social media profiles.

Recommendations:
- Review and remove sensitive personal information from public profiles
- Use unique usernames across platforms
- Enable two-factor authentication on all accounts

Remediation Checklist

Progress 0%

- [ ] **GitHub Privacy Settings** MEDIUM
  Review your GitHub account privacy settings

- [ ] **LinkedIn Privacy Settings** MEDIUM
  Review your LinkedIn account privacy settings

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THANK YOU